Tableau Training 01

1. Environment Setup

1. **Download Tableau Desktop**
    - Visit the Tableau website and download the Tableau Desktop installer.
    - Register and provide details to access the installer.
    - Students can create a Tableau account (https://www.tableau.com/en-gb/trial/tableau-software) using their personal or university email address. This account will be valid for one year.
2. **Installation**
    - Run TableauDesktop-64bit-2024-3-1.exe.
    - Accept the license agreement and install it.
3. **Verify Installation**
    - Launch Tableau from the Windows Start menu.

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2. Get Started with Tableau

1. **Connect to a Data Source**
    - Open Tableau and choose "Connect" → "Excel" → Locate *Sample – Superstore.xls*.
    - Select the **Orders** sheet.
2. **Choose Dimensions and Measures**
    - Drag dimensions like Category and Region, and measure like Sales to the worksheet.
3. **Apply Visualization**
    - Drag the SUM(Sales) column to the **Columns shelf** to create a bar chart.
    - Add another dimension to apply colours and enhance the chart.

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3. Navigation in Tableau

- **File Menu**: Create, open, and save workbooks.
- **Data Menu**: Connect, replace, or refresh data sources.
- **Worksheet Menu**: Manage views, titles, and updates.
- **Dashboard Menu**: Create dashboards, add actions, and export images.

- **Story Menu**: Create and manage sequences of worksheets or dashboards.
- **Analysis Menu**: Perform calculations, forecasts, and add trend lines.
- **Map Menu**: Work with geographic data and map layers.
- **Format Menu**: Apply borders, text formatting, and themes.
- **Server Menu**: Publish workbooks to Tableau Server.

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4. Design Flow

As Tableau helps analyse lots of data over diverse periods, dimensions, and measures, it needs meticulous planning to create a good dashboard or story. Hence, knowing the approach to design a good dashboard is essential. Like any other field of human endeavour, there are many best practices to be followed to create good worksheets and dashboards. Though the outcome expected from a Tableau project is ideally a dashboard with the story, many intermediate steps need to be completed to reach this goal. Following is a flow diagram of design steps that should be ideally followed to create effective dashboards.

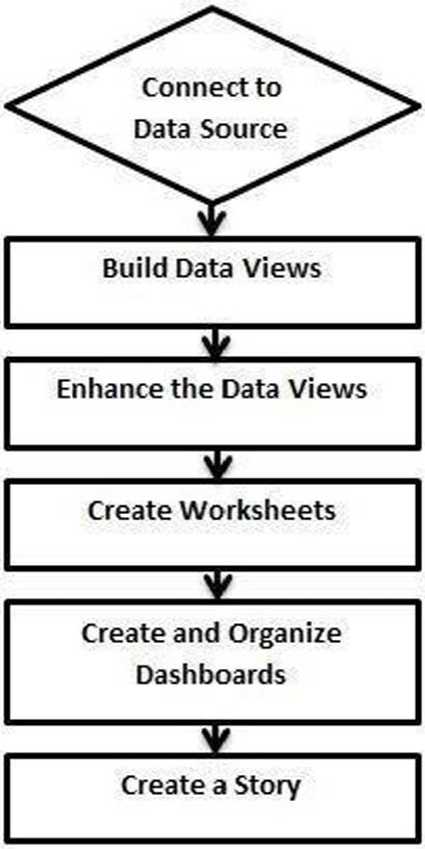


Fig 1: Design flow for a Data Analysis Process

1. **Connect to Data Source**: Use Tableau's connectors to connect to various data types.
2. **Build Data Views**: Classify data as dimensions and measures, then build reports.
3. **Enhance Views**: Apply filters, labels, and formatting.
4. **Create Worksheets**: Build different views using drag-and-drop.
5. **Organize Dashboards**: Link multiple worksheets into dashboards.
6. **Create a Story**: Combine worksheets/dashboards into an interactive narrative.

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5. File Types

- .twb: Tableau Workbook
- .twbx: Packaged Workbook (includes data)
- .tds: Data Source connection details
- .tdsx: Packaged Data Source
- .tde: Compressed Tableau Data Extract
- .tbm: Tableau Bookmark
- .tps: Tableau Preferences

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6. Show Me

1. **Analyse Two Fields**
    - Select two fields (e.g., Order Date & Profit) → Use "Show Me" → Choose **Line Chart**.
2. **Analyse Multiple Fields**
    - Select multiple fields → Tableau will grey out inappropriate charts.
    - Choose an active chart like **Scatter Plot** for multi-field visualisation.